

SAFWAN YASFEE FAHIM

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[Email](#) | [GitHub](#) | [Portfolio](#) | [LinkedIn](#)

Education

American International University-Bangladesh

Bachelor of Science in Computer Science

January 2023 –

Present

Safiuddin Sarker Academy & College, Gazipur

Higher Secondary Certificate (HSC) in Science

2021

Profile

Motivated Computer Science and Engineering student with a strong academic foundation and practical experience in data analytics, machine learning, information systems, database management, and web development. Proficient in applying technical and analytical skills to problem-solving, research, data-driven decision-making, and system development. Experienced with Python, SQL, PHP, JavaScript, C#, R, and related development and analytical tools. Possesses strong analytical thinking, problem-solving, communication, teamwork, and research skills, with a commitment to continuous learning and professional growth.

Research Experience

Research Project — Continuous-Time Modeling for Irregular SCADA Telemetry | Spring 2025-2026

- Developing a continuous-time forecasting framework for asynchronous and irregularly sampled SCADA telemetry in renewable energy grids.
- Investigating the integration of Neural Ordinary Differential Equations (Neural ODEs) and Transformer-based architectures to model continuous temporal dynamics and long-term dependencies.
- Working with solar irradiance time-series data from the NSRDB and applying self-supervised masked trajectory reconstruction for robust temporal representation learning.
- Evaluating the proposed approach against conventional time-series and deep learning methods to assess forecasting performance on irregular and incomplete data.
- Conducting literature review, data preprocessing, model development,

experimentation, and performance evaluation as part of the research work.

Research Skills: Time-Series Analysis | Machine Learning | Neural ODEs | Transformers | Self-Supervised Learning | Data Preprocessing | Python

AI-Based Student Performance Prediction and Academic Risk Analysis System | Spring 2025-2026

- Developed an ML-based system to predict student academic performance and identify academically at-risk students using academic, demographic, and behavioral data.
- Applied CRISP-DM, data preprocessing, feature engineering, class-imbalance handling, and stratified train-test validation.
- Implemented and compared Logistic Regression, Random Forest, XGBoost, CatBoost, Stacking, and Soft Voting models.
- Applied SHAP-based explainability and decision-threshold optimization for improved risk detection.
- Achieved 80% accuracy, 83% recall, 77% F1-score, and 88.7% ROC-AUC in the final system.

Projects

Dhaka Road & Draining Issue Reporting System | Spring 2025-2026

- Built responsive Next.js 14 frontend for civic infrastructure issue reporting (paired with NestJS backend API)
- Implemented JWT authentication, form validation (Zod, React Hook Form), and real-time data integration via Axios
- Applied information systems design principles to streamline real-world civic issue tracking and reporting workflows

[GitHub](#)

Analysis of Factors Influencing Video Games Success on IMDb | Fall 2025-2026

- Automated data ingestion and ETL: Created an automated web-scraping pipeline in R (rvest, stringr) to extract, clean and structure unstructured metadata from 25+ top-tier titles on IMDb.
- Exploratory Data Analysis (EDA): Developed text-cleaning routines using regular expressions (Regex) and conducted non-parametric EDA (ggplot2) to evaluate skewed distribution patterns across audience age ratings.
- Statistical Modeling & Validation: Built a Simple Linear Regression model (lm) to compare community engagement with public rating scores. The model yielded a $R^2 = 0.65$ ($p < 0.05$).
- Critical Evaluation & Reporting: Identified key dataset constraints (survivorship bias in top-tier samples) and wrote an executive-ready technical report outlining actionable future expansions, including Natural Language Processing (NLP) sentiment analysis.

Online Ticket Booking System | Fall 2025-2026

Developed a PHP and MySQL-based web application for searching available flights and booking tickets, featuring passenger selection, cabin-class selection, database-driven flight results, and MVC architecture.

Technologies:

PHP | MySQL | HTML | CSS | JavaScript | MVC | XAMPP

Analysis of Global Football Match Outcomes | Fall 2025-2026

- **Simulated Real-World Data Pipeline:** Injected controlled quality issues—including 334 missing values, duplicate entries, and negative score anomalies—into a 48,000+ row historical football dataset to simulate production-level data disruption.
- **Automated Preprocessing & Cleaning:** Developed custom R functions using tidyverse and dplyr to perform median and mode imputations, handle schema conversions, and audit dirty data.
- **Statistical Analysis & Visualization:** Conducted exploratory data analysis (EDA) using summary statistics, histograms, and boxplots to uncover scoring trends, identifying a key home-advantage bias (~1.76 home vs. ~1.18 away mean goals).
- **Data Integrity & Reporting:** Documented preprocessing trade-offs (e.g., mean vs. median imputation bias) and structured analysis-ready datasets for match outcome forecasting models.

Technical Skills

Programming Languages: Python | JavaScript (ES6) | TypeScript | C++ | Java | C# | R

Databases & Data: SQL | PostgreSQL | MySQL | MongoDB | Oracle | Microsoft SQL Server | Data Modeling | Power BI

ML & Analytics: Machine Learning | Time-Series Analysis | Neural ODEs | Transformers | Self-Supervised Learning | Scikit-learn | XGBoost

Web & Frameworks: Next.js | React | Node.js | Express | NestJS | PHP | Axios | Zod | React Hook Form

Systems & Tools: System Analysis & Design (SDLC) | Business Process Modeling (BPMN/UML/ERD) | Git | GitHub | VS Code | Postman